

ACADEMIC YEAR 2025-26

MINI PROJECT REPORT

ON

IOT BASED CAR PARKING SYSTEM

Submitted by

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MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY
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KALITHEERTHALKUPPAM, PUDUCHERRY-605107

PONDICHERRY UNIVERSITY

DEPARTMENT OF INFORMATION TECHNOLOGY

BONAFIDE CERTIFICATE

This is to certify that the Mini Project Report Titled “**IOT BASED CAR PARKING SYSTEM**” is a Bonafide record work done by **FARHANAH FARVEEN.A (24TH0214)**, **PADMASRI.M (24TH0242)**, **PRIYANGA.M (24TH0247)**, **SABIQA.M (24TH0255)** of Fourth Semester B.TECH in INFORMATION TECHNOLOGY during the academic year 2025-2026.

PROJECT GUIDE

HEAD OF THE DEPARTMENT



MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

An Autonomous Institution

Affiliated to Pondicherry University, Approved by AICTE, New Delhi
Accredited by NBA, New Delhi and NAAC With 'A' Grade
Kalitheerthalkuppam, Puducherry - 605 107



DEPARTMENT OF INFORMATION TECHNOLOGY

INSTITUTE VISION

To be a globally reputed Technical Institution creating competent leaders and Skillful innovators in Science, Technology and Management.

DEPARTMENT VISION

To transform the individuals into globally proficient Information Technologists, to meet the challenges of the evolving society.

INSTITUTE MISSION

IM1: Providing a dynamic and creative learning environment for its students to acquire exemplary technical, analytical, professional skills.

IM2: Imbibing a spirit of innovation and research among its students and faculty for solving critical problems.

IM3: Promoting Innovation, Employability and entrepreneurship skills through industry academia collaboration.

IM4: Serving the society through technical intervention and creating socially responsible Professionals.

DEPARTMENT MISSION

DM1: To provide a learner - centric environment enriched with state-of-the-art infrastructure and digital tools for providing quality education.

DM2: To encourage critical thinking, innovation and interdisciplinary research for solving real - world problems.

DM3: To strengthen industry - academia partnerships for enhancing skills, employability, and entrepreneurship.

DM4: To instill ethical values, social responsibility, and leadership qualities for nation- building and global citizenship.

Program Educational Objectives(PEO)

PEO1: Professional Competence and Employability: Demonstrate technical proficiency and problem-solving abilities to excel in the IT profession across diverse industries.

PEO2: Higher Education and Research: Pursue advanced studies, certifications, or research in emerging areas of Information Technology and allied disciplines.

PEO3: Innovation and Entrepreneurship: Initiate, manage, or contribute to technology-based start-ups or projects by applying innovative and entrepreneurial skills.

PEO4: Ethics and Social Responsibility: Exhibit ethical conduct, effective communication, teamwork, and commitment to lifelong learning for the betterment of society.



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DEPARTMENT OF INFORMATION TECHNOLOGY

Program Outcomes (POs)

- PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design / Development of Solutions:** Design creative solutions for complex engineering problems and design / develop systems / components / processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research - based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse /multi-disciplinary teams.
- PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision - making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

- PSO1: Software Design and Development:** Graduates will be able to design, develop, test, and maintain efficient and scalable software systems using appropriate computing tools and techniques.
- PSO2: Emerging Technologies and Digital Transformation:** Graduates will be able to apply knowledge of emerging technologies such as Artificial Intelligence, Cloud Computing, Cybersecurity, IoT, and Blockchain to develop sustainable and innovative IT solutions that drive digital transformation across sectors

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CHAPTER -1

ABSTRACT

In recent times, due to rapid urbanization and the increasing number of vehicles, managing parking efficiently has become a major challenge in modern cities. Traditional parking systems rely on manual supervision and cannot provide real-time information, leading to traffic congestion, wasted fuel, and frustration for drivers.

This project introduces an **“IoT-based smart car parking system”** that automates parking management and provides real-time updates on available spaces. Sensors installed in each parking slot detect vehicle presence and send the information to a centralized platform. Users can access this data through a mobile app or web dashboard, allowing them to find parking quickly, reduce search time, and avoid congestion.

By minimizing manual monitoring, improving space utilization, and providing timely information, this system offers a practical, cost-effective, and scalable solution for smart city parking management.

CHAPTER -2

OBJECTIVES OF THE MINI PROJECT

The primary objectives of IOT based car parking system are as follows:

- Monitor parking slot availability in real-time.
- Reduce time spent searching for parking.
- Optimize use of available parking spaces.
- Minimize traffic congestion inside parking areas.
- Automate parking management and reduce manual effort.
- Provide real-time updates to users via mobile app or dashboard.
- Collect data for better planning and management.
- Ensure safety and efficiency in parking operations.

CHAPTER – 3

INTRODUCTION

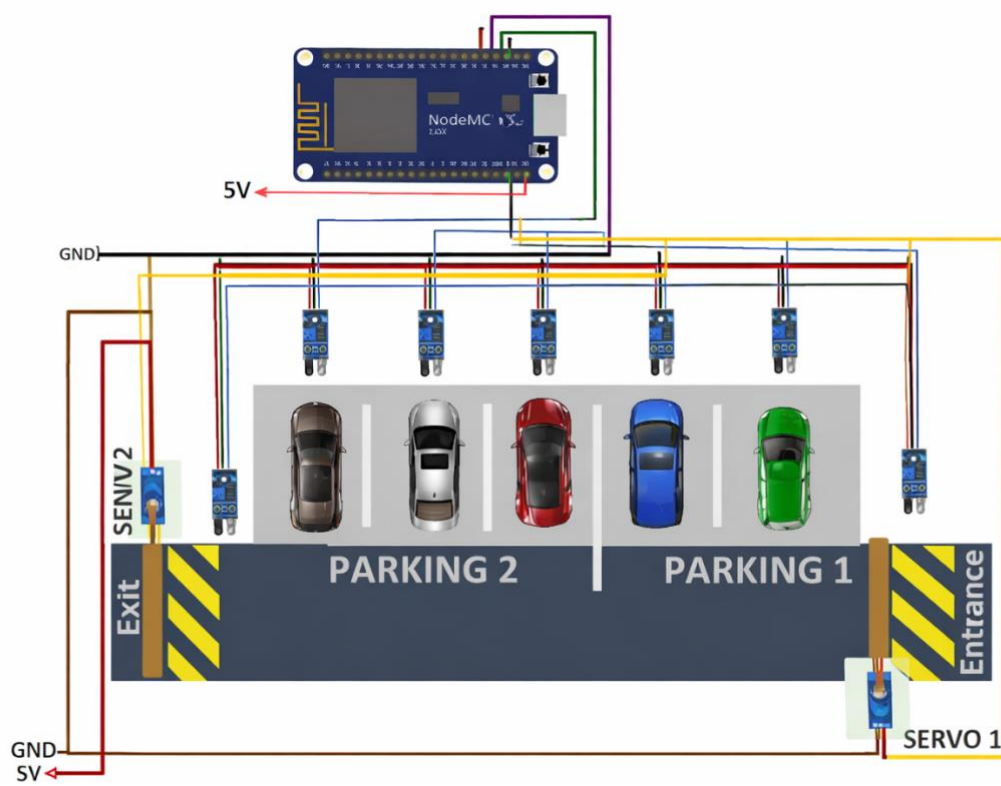
With the rapid growth of cities and increasing vehicles, finding parking has become a major problem. Drivers often spend a lot of time searching for spaces, especially in busy areas like malls, offices, hospitals, and public parking zones. This leads to traffic congestion, wasted fuel, and more pollution. Traditional parking systems rely on manual monitoring and basic infrastructure, which cannot provide real-time information and are inefficient.

To address this, we designed an **IoT-based car parking system** that automatically monitors parking spaces and provides real-time updates. Using sensors, microcontrollers, and wireless communication, the system detects available slots and sends the information to a mobile app or display, allowing users to find parking quickly.

The system saves time, reduces traffic and fuel consumption, lowers operational costs, and removes the need for manual supervision. It is a smart, cost-effective, and practical solution for modern urban parking challenges, showing how IoT can improve city life and convenience for users.

CHAPTER – 4

ARCHITECTURE DESIGN



CHAPTER – 5

SYSTEM REQUIREMENTS

Hardware Components:

1. NodeMCU (ESP8266):

Acts as the main controller. It reads data from IR sensors and sends parking status through Wi-Fi.



Features:

- Built-in Wi-Fi
- Low power consumption
- Multiple GPIO pins

Functions:

- Reads IR sensor signals
- Decides slot status (free/occupied)
- Sends data to cloud via Wi-Fi

2. IR Obstacle Sensors:

Detect the presence or absence of vehicles in each parking slot.



Features:

- Fast response
- Low cost
- Simple digital output

Functions:

- Detects vehicle presence
- Sends HIGH/LOW signal to NodeMCU

3. DC Power Adapter:

Supplies power to the entire system.



Features:

- Continuous power
- Simple connections

Functions:

- Supplies power
- Connects all components

4. Buck Converter (Voltage Regulator):

Steps down and regulates the input voltage to a safe level for NodeMCU and sensors.



Features:

- High efficiency voltage conversion
- Protects sensitive electronics
- Stable output voltage

Functions:

- Converts higher input voltage to safe level
- Supplies stable power to NodeMCU and sensors

5. Jumper Wires:

Used to connect sensors, controller, and power modules.



Features:

- Flexible and reusable
- Easy connections

Functions:

- Connect sensors, controller, and power lines

6. Mounting Board:

Holds all components firmly for proper arrangement and safety.



SOFTWARE COMPONENTS

1. Arduino IDE:

Used to write, compile, and upload the program to the NodeMCU (ESP8266).

Features:

- User-friendly interface
- Error detection and debugging
- Supports ESP8266 boards

Functions:

- Used to write, compile, and upload code
- Serial monitoring for debugging

2. ESP8266 Board Package:

This software package enables the Arduino IDE to recognize and program the ESP8266.

Functions:

- Adds ESP8266 support
- Enables Wi-Fi libraries

3. Wi-Fi Communication Libraries

These libraries enable the ESP8266 to connect to the internet.

Functions:

- Remote monitoring
- Remote fan control

4. Embedded C / Arduino C:

Programming language used for sensor reading and data processing.

Functions:

- Reads IR sensor inputs
- Implements decision logic
- Controls Wi-Fi communication

5. IoT Platform / Web Interface

Used for remote access and visualization.

Examples:

- Blynk
- MQTT
- Web Server

Functions:

- Displays temperature
- Allows manual fan control

CHAPTER – 6

IMPLEMENTATION

Step 1: Requirement Analysis

- Identify the need for a smart car parking system to reduce manual parking management.
- Define system requirements such as:
 - Number of parking slots
 - Vehicle detection using IR sensors
 - Real-time parking status
 - Internet-based monitoring
 - Select ESP8266 NodeMCU for Wi-Fi connectivity and control.

Step 2: Hardware Setup

- Place all hardware components on a flat working surface.
- Components used:
 - ESP8266 NodeMCU
 - IR obstacle sensors (for each parking slot)
 - Buck converter / power module
 - Jumper wires
 - Ensure correct power supply (5V for sensors and 3.3V logic for ESP8266).
 - Test individual components before final connections.

Step 3: Circuit Connections

- Connect IR sensor VCC pins to 5V supply.
- Connect IR sensor GND pins to common ground.
- Connect IR sensor OUT pins to ESP8266 GPIO pins.
- Connect buck converter to provide regulated power supply.
- Ensure common ground between ESP8266, sensors, and power module.

Step 4: Software Installation

- Install Arduino IDE on the computer.
- Add ESP8266 Board Manager in Arduino IDE.
- Install required ESP8266 Wi-Fi libraries.

- Select correct board (NodeMCU ESP8266) and COM port.

Step 5: Program Development

- Write program to:
 - Read vehicle detection status from IR sensors.
 - Identify whether parking slots are occupied or vacant.
 - Count available parking spaces.
 - Send parking status to IoT cloud/server using Wi-Fi.
 - Update parking availability in real time.

Step 6: Uploading the Program

- Connect ESP8266 NodeMCU to the computer using USB cable.
- Compile and upload the program through Arduino IDE.
- Use Serial Monitor to verify sensor readings and Wi-Fi connection.

Step 7: Testing and Calibration

- Power ON the system.
- Place an object or vehicle in front of IR sensors.
- Observe slot status change (occupied / vacant).
- Adjust IR sensor sensitivity using onboard potentiometer.
- Verify correct data transmission over Wi-Fi.

Step 8: Integration with IoT Platform

- Configure IoT platform or web dashboard.
- Display real-time parking slot availability.
- Enable remote monitoring using mobile or web application.
- Update parking status automatically when vehicle position changes.

Step 9: Final Deployment

- Fix sensors properly at each parking slot.
- Mount ESP8266 and power module securely.
- Ensure stable power supply and Wi-Fi connectivity.
- Deploy system for real-time smart parking management.

CHAPTER – 7

SOURCE CODE

```
#define BLYNK_TEMPLATE_ID "TMPLxxxx"

#define BLYNK_TEMPLATE_NAME "Smart Parking"

#define BLYNK_AUTH_TOKEN "Your_Blynk_Auth_Token"

#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

char ssid[] = "vivo T3x 5G";

char pass[] = "padma2007";

LiquidCrystal_I2C lcd(0x27, 16, 2);

// IR sensor pins

#define IR1 D1

#define IR2 D2

#define IR3 D5

#define IR4 D6

void setup()

{

  Serial.begin(9600);

  pinMode(IR1, INPUT);

  pinMode(IR2, INPUT);

  pinMode(IR3, INPUT);

  pinMode(IR4, INPUT);

  lcd.begin();

  lcd.backlight();

  lcd.setCursor(0,0);

  lcd.print("Car Parking IoT");

  delay(2000);
```

```
lcd.clear();

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

}

void loop()

{

Blynk.run();

  int s1 = digitalRead(IR1);

  int s2 = digitalRead(IR2);

  int s3 = digitalRead(IR3);

  int s4 = digitalRead(IR4);

lcd.setCursor(0,0);

lcd.print("S1:");

lcd.print(s1 == LOW ? "Occ" : "Emp ");

lcd.print("S2:");

lcd.print(s2 == LOW ? "Occ" : "Emp");

lcd.setCursor(0,1);

lcd.print("S3:");

lcd.print(s3 == LOW ? "Occ" : "Emp ");

lcd.print("S4:");

lcd.print(s4 == LOW ? "Occ" : "Emp");

  // Blynk LEDs

Blynk.virtualWrite(V0, s1 == LOW ? 255 : 0);

Blynk.virtualWrite(V1, s2 == LOW ? 255 : 0);

Blynk.virtualWrite(V2, s3 == LOW ? 255 : 0);

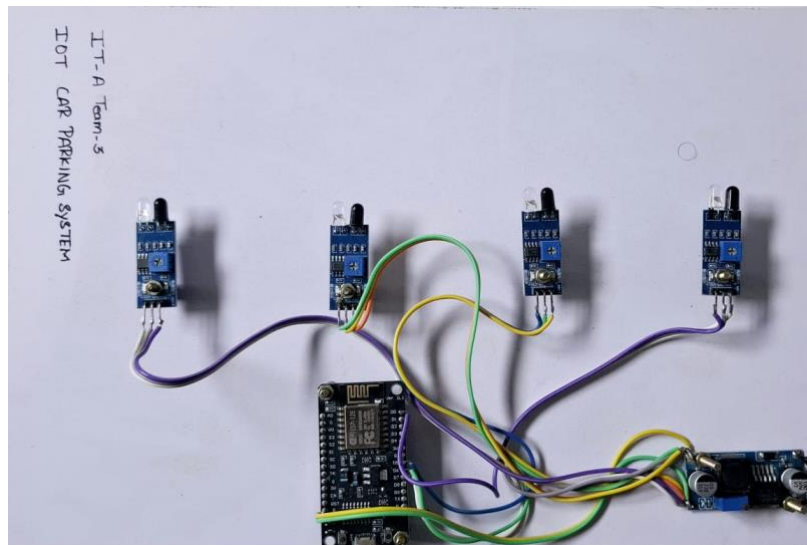
Blynk.virtualWrite(V3, s4 == LOW ? 255 : 0);

delay(500);

}
```

CHAPTER –8

OUTPUT



CHAPTER – 9

CONCLUSION

The IoT-based Car Parking System offers an intelligent solution for modern parking challenges. By using sensors and real-time data, it accurately detects and displays available parking spaces, allowing drivers to locate spots quickly and efficiently. This reduces traffic congestion, saves time, and enhances convenience for users. Additionally, it supports better management of parking areas for authorities, making the system both practical and effective. Overall, the project demonstrates how IoT technology can improve urban infrastructure and everyday life.